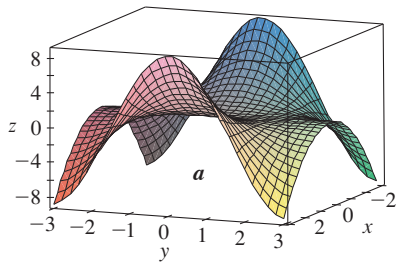
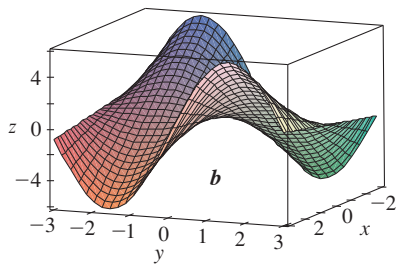
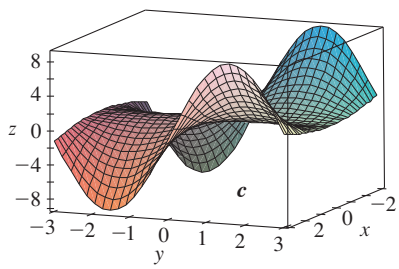
Suggested problems: 1-39 odd

Problem 1: The following surfaces, labeled a, b, and c, are graphs of a function f and its partial derivatives f_x and f_y . Identify each surface and give reasons for your choices.







We have $c = f$, $a = f_y$, and $b = f_x$.

Problem 2: The temperature at a point (x, y) on a flat metal plate is given by

$$T(x, y) = \frac{60}{1 + x^2 + y^2},$$

where T is measured in °C and x and y in meters. Find the rate of change of temperature with respect to distance at the point $(2, 1)$ in the x-direction and the y-direction.

We have

$$T_x(x, y) = -\frac{120x}{(1 + x^2 + y^2)^2} \quad \text{and} \quad T_y(x, y) = -\frac{120y}{(1 + x^2 + y^2)^2}.$$

The rate of change in the x-direction is then

$$T_x(2, 1) = -20/3 \approx -6.66 \text{ °C per meter}$$

and in the y-direction

$$T_y(2, 1) = -10/3 \approx -3.33 \text{ °C per meter.}$$